

Project 1: Punta Cana Airbnb Pricing Analysis

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04-19-2026

Introduction

The purpose of this project is to analyze Airbnb listings in Punta Cana, Dominican Republic, and determine which listing characteristics are associated with higher nightly rental rates. Punta Cana is a major tourism destination, and understanding price drivers can help hosts optimize pricing strategies and help guests understand what influences cost.

This project follows the structure required in the assignment:

- A clear research question
- At least two pages of exploratory data analysis (EDA)
- At least three visualizations
- A multiple regression model
- Interpretation of results
- Limitations and conclusions

Research question:

Which listing characteristics best explain variation in nightly Airbnb prices in Punta Cana?

To answer this question, I followed a clear, step-by-step process:

1. Load and inspect the raw dataset
2. Clean and wrangle the data
3. Identify numeric variables suitable for modeling
4. Perform exploratory data analysis (EDA)

5. Select predictors of price
6. Fit a multiple regression model
7. Interpret the model and address overlapping predictors
8. Discuss limitations and conclusions

Step 1: Dataset Description and Loading

The dataset comes from AirRoi.com (exported April 4, 2026). It contains Airbnb listings in Punta Cana with variables describing capacity and pricing.

Key variables include:

- **ttm_avg_rate**: trailing twelve-month average nightly rate
- **guests**: maximum guest capacity
- **bedrooms**: number of bedrooms
- **baths**: number of bathrooms

In this analysis, `ttm_avg_rate` is renamed to `price` for clarity.

```
airbnb <- read_csv("PuntaCanaListingsData-AirRoi.com-04-04-2026.csv") |>
  mutate(
    guests = as.numeric(guests),
    bedrooms = as.numeric(bedrooms),
    baths = as.numeric(baths),
    price = ttm_avg_rate
  )

glimpse(airbnb)
```

```
## Rows: 195
## Columns: 63
## $ listing_id      <dbl> 2.877218e+07, 4.509088e+07, 7.252100e+17, ~
## $ listing_name    <chr> "FULL FURNISHED ONE ROOM APARTMENT", "Brea~
## $ listing_type     <chr> "Entire condo", "Entire condo", "Entire vi~
## $ room_type        <chr> "entire_home", "entire_home", "entire_home~
## $ cover_photo_url  <chr> "https://a0.muscache.com/im/pictures/miso/~
## $ photos_count     <dbl> 25, 29, 43, 75, 40, 54, 90, 44, 47, 31, 24~
## $ host_id          <dbl> 1173370, 246641852, 442609295, 430645153, ~
## $ host_name        <chr> "Juan", "Beyond Punta Cana", "Miguel", "Li~
## $ cohost_ids        <dbl> NA, NA, NA, 4.755124e+07, NA, NA, NA, NA, ~
## $ cohost_names      <chr> NA, NA, NA, "Tatiana", NA, NA, NA, NA, NA,~
## $ superhost         <lgl> TRUE, FALSE, FALSE, TRUE, FALSE, FALSE, FA~
## $ latitude          <dbl> 18.6848, 18.5040, 18.4815, 18.4989, 18.670~
## $ longitude         <dbl> -68.4557, -68.3870, -68.4085, -68.3808, -6~
## $ guests            <dbl> 2, 6, 16, 4, 3, 6, 14, 4, 6, 6, 3, 4, 9, 6~
## $ bedrooms          <dbl> 1, 2, 6, 2, 1, 3, 5, 2, 2, 3, 1, 1, 4, 3, ~
## $ beds             <dbl> 1, 2, 9, 2, 1, 4, 8, 2, 3, 4, 1, 1, 6, 2, ~
## $ baths             <dbl> 1.0, 2.0, 5.5, 2.0, 1.0, 2.5, 5.5, 3.0, 2.~
## $ registration      <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, ~
```

## \$ amenities	<chr> "TV,Window guards,Wifi,Air conditioning,Sk~
## \$ instant_book	<lgl> FALSE, TRUE, NA, NA, TRUE, NA, FALSE, NA, ~
## \$ professional_management	<lgl> FALSE, FALSE, FALSE, FALSE, NA, FALSE, FAL~
## \$ min_nights	<dbl> NA, NA, NA, NA, NA, NA, 3, NA, NA, NA, NA,~
## \$ cancellation_policy	<chr> "Moderate", "Firm", "Firm", "Firm", "Flexi~
## \$ currency	<chr> "DOP", "DOP", "DOP", "DOP", "DOP", "DOP", ~
## \$ cleaning_fee	<dbl> 18, 60, 0, 125, 60, 100, 70, 35, 45, 65, 4~
## \$ extra_guest_fee	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## \$ num_reviews	<dbl> 5, 8, 7, 13, 1, 7, 1, 8, 9, 15, 17, 25, 30~
## \$ rating_overall	<dbl> 4.80, 4.88, 4.86, 5.00, NA, 4.71, NA, 4.88~
## \$ rating_accuracy	<dbl> 5.0, 4.9, 4.9, 4.9, NA, 4.6, NA, 4.9, 4.8,~
## \$ rating_checkin	<dbl> 5.0, 5.0, 4.9, 5.0, NA, 4.7, NA, 5.0, 4.9,~
## \$ rating_cleanliness	<dbl> 4.8, 4.8, 5.0, 4.9, NA, 4.4, NA, 4.9, 4.8,~
## \$ rating_communication	<dbl> 5.0, 5.0, 4.7, 4.9, NA, 4.9, NA, 4.9, 4.9,~
## \$ rating_location	<dbl> 5.0, 5.0, 5.0, 4.8, NA, 4.9, NA, 4.9, 4.9,~
## \$ rating_value	<dbl> 4.8, 4.8, 4.9, 5.0, NA, 4.6, NA, 4.9, 5.0,~
## \$ ttm_revenue	<dbl> 908, 17195, 75577, 35705, 3094, 8358, 1164~
## \$ ttm_revenue_native	<dbl> 53505, 1013235, 4452914, 2103839, 182334, ~
## \$ ttm_avg_rate	<dbl> 26.5, 306.1, 2181.6, 257.8, 89.4, 119.7, 8~
## \$ ttm_avg_rate_native	<dbl> 1583.1, 18285.0, 130500.5, 15401.1, 5338.9~
## \$ ttm_occupancy	<dbl> 0.090, 0.159, 0.099, 0.342, 0.099, 0.175, ~
## \$ ttm_adjusted_occupancy	<dbl> 0.101, 0.183, 0.000, 0.352, 0.000, 0.204, ~
## \$ ttm_revpar	<dbl> 2.4, 44.5, 207.1, 90.3, 8.1, 20.4, 31.5, 6~
## \$ ttm_revpar_native	<dbl> 143.7, 2621.0, 12199.8, 5320.0, 480.2, 120~
## \$ ttm_adjusted_revpar	<dbl> 2.7, 51.2, 0.0, 92.8, 0.0, 23.8, 35.7, 0.0~
## \$ ttm_adjusted_revpar_native	<dbl> 160.9, 3017.9, 0.0, 5469.9, 0.0, 1399.5, 2~
## \$ ttm_reserved_days	<dbl> 33, 58, 36, 125, 36, 64, 21, 18, 42, 93, 4~
## \$ ttm_blocked_days	<dbl> 39, 48, 0, 10, 0, 51, 43, 0, 0, 0, 0, 0, 0~
## \$ ttm_available_days	<dbl> 332, 307, 329, 240, 329, 301, 344, 347, 32~
## \$ ttm_total_days	<dbl> 365, 365, 365, 365, 365, 365, 365, 365, 36~
## \$ 190d_revenue	<dbl> 0, 1688, 45824, 15580, 0, 2394, 0, 0, 0, 1~
## \$ 190d_revenue_native	<dbl> 0, 99455, 2699900, 917956, 0, 141054, 0, 0~
## \$ 190d_avg_rate	<dbl> 25.9, 370.6, 2258.8, 352.0, 91.4, 133.3, 8~
## \$ 190d_avg_rate_native	<dbl> 1526.8, 21837.3, 133084.1, 20742.0, 5387.7~
## \$ 190d_occupancy	<dbl> 0.000, 0.056, 0.244, 0.444, 0.000, 0.178, ~
## \$ 190d_adjusted_occupancy	<dbl> 0.000, 0.119, 0.000, 0.500, 0.000, 0.410, ~
## \$ 190d_revpar	<dbl> 0.0, 18.1, 509.2, 156.4, 0.0, 23.3, 0.0, 0~
## \$ 190d_revpar_native	<dbl> 0.0, 1065.8, 29998.9, 9217.5, 0.0, 1370.9,~
## \$ 190d_adjusted_revpar	<dbl> 0.0, 38.8, 0.0, 176.0, 0.0, 53.7, 0.0, 0.0~
## \$ 190d_adjusted_revpar_native	<dbl> 0.0, 2283.8, 0.0, 10369.7, 0.0, 3163.5, 0.~
## \$ 190d_reserved_days	<dbl> 0, 5, 22, 40, 0, 16, 0, 0, 0, 42, 17, 21, ~
## \$ 190d_blocked_days	<dbl> 0, 48, 0, 10, 0, 51, 0, 0, 0, 0, 0, 0, 0, ~
## \$ 190d_available_days	<dbl> 90, 85, 68, 50, 90, 74, 90, 90, 90, 48, 73~
## \$ 190d_total_days	<dbl> 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90~
## \$ price	<dbl> 26.5, 306.1, 2181.6, 257.8, 89.4, 119.7, 8~

Step 2: Data Cleaning and Wrangling

```
airbnb <- airbnb |>
  filter(!is.na(price), price > 0) |>
  filter(!is.na(guests), !is.na.bedrooms), !is.na(baths))

summary(airbnb)
```

```
##      listing_id      listing_name      listing_type      room_type
## Min.      :2.877e+07      Length:190      Length:190      Length:190
## 1st Qu.:1.153e+18      Class :character      Class :character      Class :character
## Median :1.255e+18      Mode  :character      Mode  :character      Mode  :character
## Mean      :1.199e+18
## 3rd Qu.:1.321e+18
## Max.      :1.337e+18
##
##      cover_photo_url      photos_count      host_id      host_name
## Length:190      Min.      : 9.00      Min.      : 1173370      Length:190
## Class :character      1st Qu.: 23.00      1st Qu.:305565439      Class :character
## Mode  :character      Median : 40.00      Median :558880554      Mode  :character
##
##      Mean      : 57.37      Mean      :445797578
##      3rd Qu.: 63.50      3rd Qu.:581682514
##      Max.      :245.00      Max.      :672093916
##
##      cohost_ids      cohost_names      superhost      latitude
## Min.      :2.052e+07      Length:190      Mode :logical      Min.      :18.46
## 1st Qu.:4.764e+08      Class :character      FALSE:111      1st Qu.:18.60
## Median :4.764e+08      Mode  :character      TRUE :79      Median :18.64
## Mean      :3.052e+33
##      3rd Qu.:6.709e+08      Mean      :18.65
##      Max.      :2.157e+35      3rd Qu.:18.69
##      NA's      :117      Max.      :18.85
##
##      longitude      guests      bedrooms      beds
## Min.      :-68.62      Min.      : 1.000      Min.      :1.000      Min.      : 1.000
## 1st Qu.: -68.43      1st Qu.: 3.000      1st Qu.:1.000      1st Qu.: 2.000
## Median : -68.41      Median : 4.000      Median :2.000      Median : 2.500
## Mean      : -68.44      Mean      : 4.884      Mean      :2.147      Mean      : 2.753
## 3rd Qu.: -68.40      3rd Qu.: 6.000      3rd Qu.:3.000      3rd Qu.: 3.000
## Max.      : -68.34      Max.      :16.000      Max.      :7.000      Max.      :12.000
##
##      baths      registration      amenities      instant_book
## Min.      :1.00      Mode :logical      Length:190      Mode :logical
## 1st Qu.:1.00      FALSE:190      Class :character      FALSE:14
## Median :2.00      Mode  :character      TRUE :7
## Mean      :2.15
##      3rd Qu.:2.50      NA's :169
```

```

## Max.      :7.50
##
## professional_management  min_nights  cancellation_policy  currency
## Mode :logical          Min.   :2.00    Length:190          Length:190
## FALSE:180              1st Qu.:2.25    Class :character    Class :character
## TRUE :4                 Median :2.50    Mode  :character    Mode  :character
## NA's :6                 Mean    :2.50
##                        3rd Qu.:2.75
##                        Max.    :3.00
##                        NA's    :188
## cleaning_fee  extra_guest_fee  num_reviews  rating_overall
## Min.   : 0.00  Min.   : 0.000  Min.   : 1.00  Min.   :3.00
## 1st Qu.:10.00  1st Qu.: 0.000  1st Qu.: 7.00  1st Qu.:4.60
## Median :35.00  Median : 0.000  Median :15.00  Median :4.80
## Mean   :40.01  Mean   : 1.085  Mean   :22.03  Mean   :4.71
## 3rd Qu.:50.00  3rd Qu.: 0.000  3rd Qu.:28.75  3rd Qu.:4.92
## Max.   :335.00  Max.   :75.000  Max.   :160.00  Max.   :5.00
## NA's   :1      NA's   :1              NA's   :13
## rating_accuracy rating_checkin rating_cleanliness rating_communication
## Min.   :3.000  Min.   :3.400  Min.   :3.30  Min.   :3.300
## 1st Qu.:4.700  1st Qu.:4.700  1st Qu.:4.70  1st Qu.:4.700
## Median :4.800  Median :4.800  Median :4.80  Median :4.900
## Mean   :4.752  Mean   :4.766  Mean   :4.75  Mean   :4.782
## 3rd Qu.:5.000  3rd Qu.:5.000  3rd Qu.:4.90  3rd Qu.:5.000
## Max.   :5.000  Max.   :5.000  Max.   :5.00  Max.   :5.000
## NA's   :13     NA's   :13     NA's   :13     NA's   :13
## rating_location rating_value  ttm_revenue  ttm_revenue_native
## Min.   :3.700  Min.   :3.300  Min.   : 212  Min.   : 12496
## 1st Qu.:4.600  1st Qu.:4.600  1st Qu.: 1970  1st Qu.: 116062
## Median :4.800  Median :4.800  Median : 4348  Median : 256154
## Mean   :4.731  Mean   :4.727  Mean   : 8225  Mean   : 484910
## 3rd Qu.:4.900  3rd Qu.:4.900  3rd Qu.: 8982  3rd Qu.: 529246
## Max.   :5.000  Max.   :5.000  Max.   :130334  Max.   :7679674
## NA's   :13     NA's   :13
## ttm_avg_rate  ttm_avg_rate_native ttm_occupancy  ttm_adjusted_occupancy
## Min.   : 18.30  Min.   : 1079  Min.   :0.0270  Min.   :0.00000
## 1st Qu.: 63.20  1st Qu.: 3760  1st Qu.:0.0550  1st Qu.:0.00000
## Median : 99.25  Median : 5896  Median :0.1055  Median :0.00000
## Mean   :177.45  Mean   :10543  Mean   :0.1346  Mean   :0.06591
## 3rd Qu.:164.88  3rd Qu.: 9792  3rd Qu.:0.1943  3rd Qu.:0.10175
## Max.   :2738.10  Max.   :161328  Max.   :0.4960  Max.   :0.48300
##
## ttm_revpar  ttm_revpar_native ttm_adjusted_revpar
## Min.   : 0.600  Min.   : 35.3  Min.   : 0.00
## 1st Qu.: 5.125  1st Qu.: 304.2  1st Qu.: 0.00
## Median :11.250  Median : 662.5  Median : 0.00
## Mean   :21.538  Mean   :1269.9  Mean   :11.85
## 3rd Qu.:23.075  3rd Qu.:1358.0  3rd Qu.:10.45

```

```

## Max.      :357.100    Max.      :21040.2    Max.      :415.10
##
## ttm_adjusted_revpar_native ttm_reserved_days ttm_blocked_days
## Min.      :    0.0          Min.      : 10.00    Min.      :    0.00
## 1st Qu.:    0.0          1st Qu.: 20.00    1st Qu.:    0.00
## Median :    0.0          Median : 38.50    Median :    0.00
## Mean      : 699.1          Mean      : 49.11    Mean      : 12.46
## 3rd Qu.: 616.5          3rd Qu.: 71.00    3rd Qu.: 15.50
## Max.      :24457.6        Max.      :181.00    Max.      :169.00
##
## ttm_available_days ttm_total_days    190d_revenue    190d_revenue_native
## Min.      :184.0          Min.      :343.0    Min.      :    0    Min.      :    0
## 1st Qu.:294.0          1st Qu.:365.0    1st Qu.: 1040    1st Qu.: 61247
## Median :326.5          Median :365.0    Median : 2186    Median : 128796
## Mean      :315.5          Mean      :364.6    Mean      : 4331    Mean      : 255152
## 3rd Qu.:344.8          3rd Qu.:365.0    3rd Qu.: 4588    3rd Qu.: 270334
## Max.      :355.0          Max.      :365.0    Max.      :49755    Max.      :2931509
##
## 190d_avg_rate    190d_avg_rate_native 190d_occupancy
## Min.      :    0.00    Min.      :    0    Min.      :0.0000
## 1st Qu.: 65.05    1st Qu.: 3835    1st Qu.:0.1330
## Median : 101.65    Median : 5991    Median :0.2110
## Mean      : 195.45    Mean      : 11516    Mean      :0.2607
## 3rd Qu.: 192.88    3rd Qu.: 11362    3rd Qu.:0.3890
## Max.      :2933.80    Max.      :172856    Max.      :0.8330
##
## 190d_adjusted_occupancy 190d_revpar    190d_revpar_native
## Min.      :0.0000    Min.      :    0.00    Min.      :    0.0
## 1st Qu.:0.0000    1st Qu.: 11.07    1st Qu.: 653.8
## Median :0.0000    Median : 23.25    Median : 1368.6
## Mean      :0.1134    Mean      : 46.24    Mean      : 2724.6
## 3rd Qu.:0.2225    3rd Qu.: 47.75    3rd Qu.: 2812.1
## Max.      :0.8490    Max.      :552.80    Max.      :32572.3
##
## 190d_adjusted_revpar 190d_adjusted_revpar_native 190d_reserved_days
## Min.      :    0.00    Min.      :    0    Min.      : 0.00
## 1st Qu.:    0.00    1st Qu.:    0    1st Qu.:12.00
## Median :    0.00    Median :    0    Median :19.00
## Mean      : 24.42    Mean      : 1439    Mean      :23.47
## 3rd Qu.: 22.60    3rd Qu.: 1330    3rd Qu.:35.00
## Max.      :1081.60    Max.      :63729    Max.      :75.00
##
## 190d_blocked_days 190d_available_days 190d_total_days    price
## Min.      : 0.000    Min.      :15.00    Min.      :90    Min.      : 18.30
## 1st Qu.: 0.000    1st Qu.:55.00    1st Qu.:90    1st Qu.: 63.20
## Median : 0.000    Median :71.00    Median :90    Median : 99.25
## Mean      : 7.989    Mean      :66.53    Mean      :90    Mean      : 177.45
## 3rd Qu.: 9.000    3rd Qu.:78.00    3rd Qu.:90    3rd Qu.: 164.88

```

```
## Max. :90.000 Max. :90.00 Max. :90 Max. :2738.10
##
```

Commentary:

- Removed listings with missing or zero prices.
- Removed listings missing guests, bedrooms, or baths.
- Ensures a clean dataset suitable for regression.

Step 3: Identifying Numeric Variables

```
str(airbnb)
```

```
## tibble [190 x 63] (S3: tbl_df/tbl/data.frame)
## $ listing_id          : num [1:190] 2.88e+07 4.51e+07 7.25e+17 7.59e+17 7.64e+17 ..
## $ listing_name        : chr [1:190] "FULL FURNISHED ONE ROOM APARTMENT" "Breathtaking" ..
## $ listing_type        : chr [1:190] "Entire condo" "Entire condo" "Entire villa" "Entire villa" ..
## $ room_type           : chr [1:190] "entire_home" "entire_home" "entire_home" "entire_home" ..
## $ cover_photo_url     : chr [1:190] "https://a0.muscache.com/im/pictures/miso/Hosting" ..
## $ photos_count        : num [1:190] 25 29 43 75 40 54 90 44 47 31 ...
## $ host_id             : num [1:190] 1.17e+06 2.47e+08 4.43e+08 4.31e+08 1.62e+07 ..
## $ host_name           : chr [1:190] "Juan" "Beyond Punta Cana" "Miguel" "Lisbeth" ..
## $ cohost_ids          : num [1:190] NA NA NA 47551244 NA ...
## $ cohost_names        : chr [1:190] NA NA NA "Tatiana" ...
## $ superhost           : logi [1:190] TRUE FALSE FALSE TRUE FALSE FALSE ...
## $ latitude            : num [1:190] 18.7 18.5 18.5 18.5 18.7 ...
## $ longitude           : num [1:190] -68.5 -68.4 -68.4 -68.4 -68.4 ...
## $ guests              : num [1:190] 2 6 16 4 3 6 14 4 6 6 ...
## $ bedrooms            : num [1:190] 1 2 6 2 1 3 5 2 2 3 ...
## $ beds               : num [1:190] 1 2 9 2 1 4 8 2 3 4 ...
## $ baths               : num [1:190] 1 2 5.5 2 1 2.5 5.5 3 2 3.5 ...
## $ registration        : logi [1:190] FALSE FALSE FALSE FALSE FALSE FALSE ...
## $ amenities           : chr [1:190] "TV,Window guards,Wifi,Air conditioning,Ski-in/Ski-out" ..
## $ instant_book        : logi [1:190] FALSE TRUE NA NA TRUE NA ...
## $ professional_management : logi [1:190] FALSE FALSE FALSE FALSE NA FALSE ...
## $ min_nights          : num [1:190] NA NA NA NA NA NA 3 NA NA NA ...
## $ cancellation_policy : chr [1:190] "Moderate" "Firm" "Firm" "Firm" ...
## $ currency            : chr [1:190] "DOP" "DOP" "DOP" "DOP" ...
## $ cleaning_fee        : num [1:190] 18 60 0 125 60 100 70 35 45 65 ...
## $ extra_guest_fee     : num [1:190] 0 0 0 0 0 0 0 0 0 0 ...
## $ num_reviews         : num [1:190] 5 8 7 13 1 7 1 8 9 15 ...
## $ rating_overall       : num [1:190] 4.8 4.88 4.86 5 NA 4.71 NA 4.88 4.89 5 ...
## $ rating_accuracy      : num [1:190] 5 4.9 4.9 4.9 NA 4.6 NA 4.9 4.8 5 ...
## $ rating_checkin       : num [1:190] 5 5 4.9 5 NA 4.7 NA 5 4.9 5 ...
## $ rating_cleanliness   : num [1:190] 4.8 4.8 5 4.9 NA 4.4 NA 4.9 4.8 4.9 ...
## $ rating_communication : num [1:190] 5 5 4.7 4.9 NA 4.9 NA 4.9 4.9 4.9 ...
## $ rating_location      : num [1:190] 5 5 5 4.8 NA 4.9 NA 4.9 4.9 5 ...
## $ rating_value         : num [1:190] 4.8 4.8 4.9 5 NA 4.6 NA 4.9 5 4.9 ...
## $ ttm_revenue         : num [1:190] 908 17195 75577 35705 3094 ...
## $ ttm_revenue_native   : num [1:190] 53505 1013235 4452914 2103839 182334 ...
## $ ttm_avg_rate         : num [1:190] 26.5 306.1 2181.6 257.8 89.4 ...
## $ ttm_avg_rate_native  : num [1:190] 1583 18285 130501 15401 5339 ...
## $ ttm_occupancy        : num [1:190] 0.09 0.159 0.099 0.342 0.099 0.175 0.058 0.049 ...
## $ ttm_adjusted_occupancy : num [1:190] 0.101 0.183 0 0.352 0 0.204 0.065 0 0 0 ...
## $ ttm_revpar           : num [1:190] 2.4 44.5 207.1 90.3 8.1 ...
```

```

## $ ttm_revpar_native      : num [1:190] 144 2621 12200 5320 480 ...
## $ ttm_adjusted_revpar    : num [1:190] 2.7 51.2 0 92.8 0 23.8 35.7 0 0 0 ...
## $ ttm_adjusted_revpar_native : num [1:190] 161 3018 0 5470 0 ...
## $ ttm_reserved_days      : num [1:190] 33 58 36 125 36 64 21 18 42 93 ...
## $ ttm_blocked_days       : num [1:190] 39 48 0 10 0 51 43 0 0 0 ...
## $ ttm_available_days     : num [1:190] 332 307 329 240 329 301 344 347 323 272 ...
## $ ttm_total_days         : num [1:190] 365 365 365 365 365 365 365 365 365 365 ...
## $ l90d_revenue           : num [1:190] 0 1688 45824 15580 0 ...
## $ l90d_revenue_native    : num [1:190] 0 99455 2699900 917956 0 ...
## $ l90d_avg_rate          : num [1:190] 25.9 370.6 2258.8 352 91.4 ...
## $ l90d_avg_rate_native   : num [1:190] 1527 21837 133084 20742 5388 ...
## $ l90d_occupancy         : num [1:190] 0 0.056 0.244 0.444 0 0.178 0 0 0 0.467 ...
## $ l90d_adjusted_occupancy : num [1:190] 0 0.119 0 0.5 0 0.41 0 0 0 0 ...
## $ l90d_revpar            : num [1:190] 0 18.1 509.2 156.4 0 ...
## $ l90d_revpar_native     : num [1:190] 0 1066 29999 9218 0 ...
## $ l90d_adjusted_revpar   : num [1:190] 0 38.8 0 176 0 53.7 0 0 0 0 ...
## $ l90d_adjusted_revpar_native : num [1:190] 0 2284 0 10370 0 ...
## $ l90d_reserved_days     : num [1:190] 0 5 22 40 0 16 0 0 0 42 ...
## $ l90d_blocked_days      : num [1:190] 0 48 0 10 0 51 0 0 0 0 ...
## $ l90d_available_days    : num [1:190] 90 85 68 50 90 74 90 90 90 48 ...
## $ l90d_total_days        : num [1:190] 90 90 90 90 90 90 90 90 90 90 ...
## $ price                  : num [1:190] 26.5 306.1 2181.6 257.8 89.4 ...

```

The numeric variables relevant to this analysis are:

- **price**
- **guests**
- **bedrooms**
- **baths**

These are the variables used for correlation analysis and regression.

Step 4: Exploratory Data Analysis (EDA)

Summary Statistics for Nightly Rate

```
summary(airbnb$price)
```

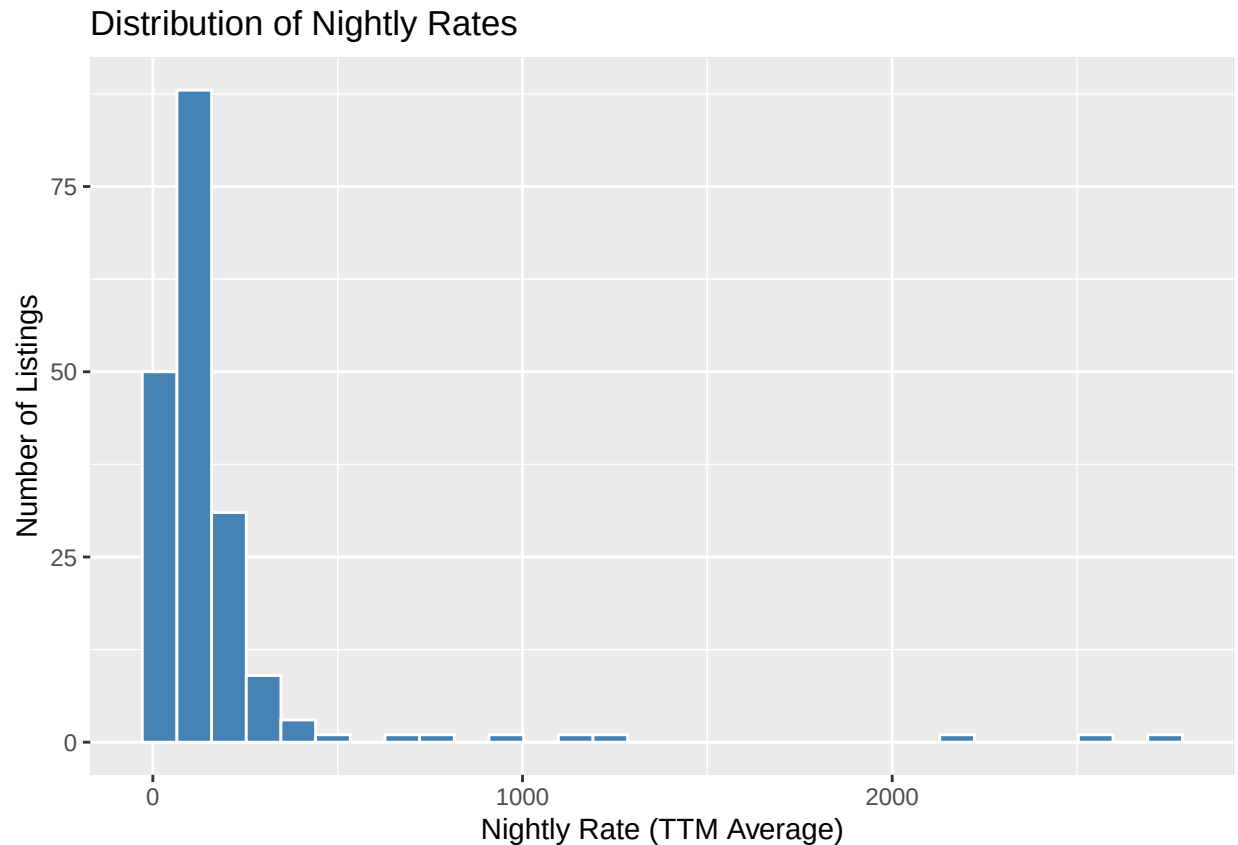
```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##   18.30   63.20   99.25  177.45  164.88 2738.10
```

Commentary:

Nightly rates are right-skewed, with most listings in the lower-to-mid range and a long tail of high-priced properties.

Distribution of Nightly Rates

```
ggplot(airbnb, aes(price)) +  
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +  
  labs(title = "Distribution of Nightly Rates",  
        x = "Nightly Rate (TTM Average)",  
        y = "Number of Listings")
```

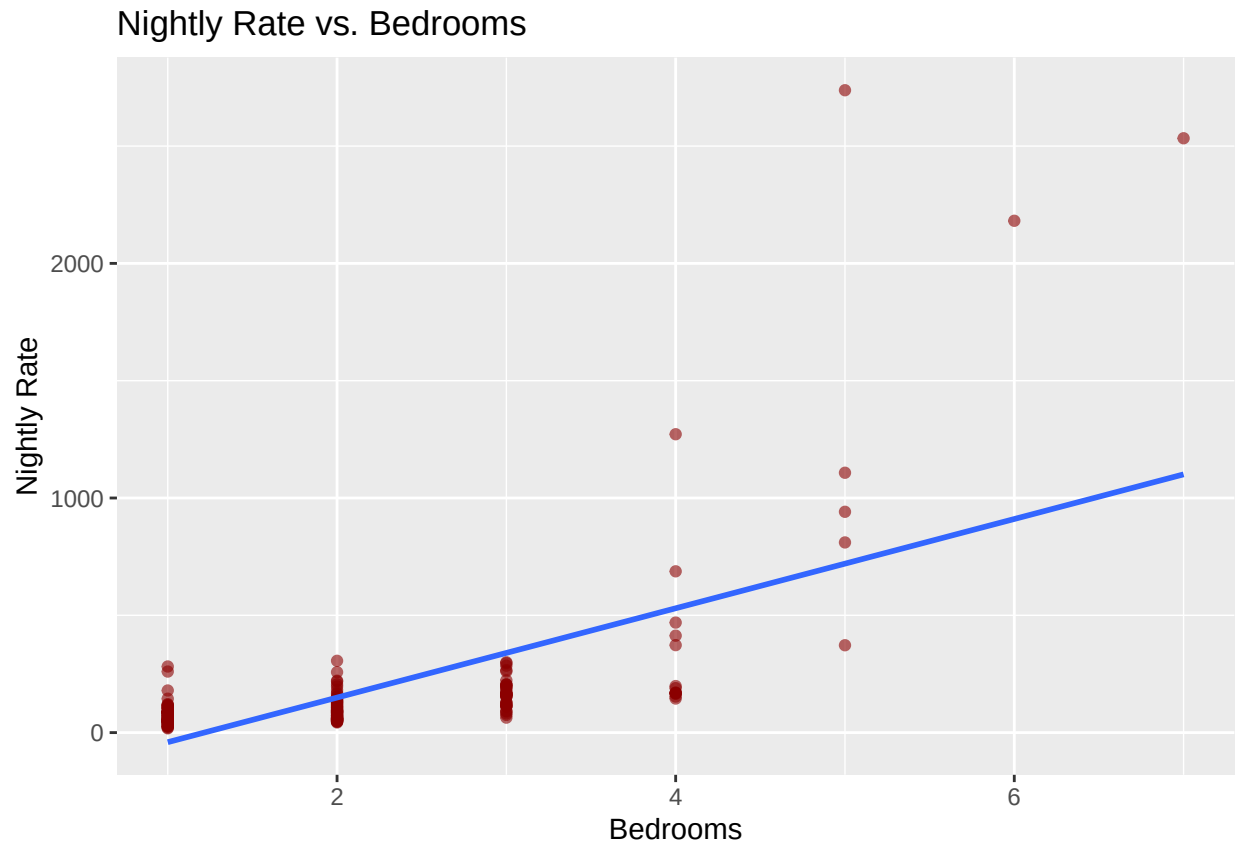


Commentary:

This histogram shows the distribution of nightly prices. The distribution is right-skewed, meaning most listings fall in the lower-to-mid price range while a smaller number of high-priced luxury listings create a long right tail. This confirms that price varies widely across Punta Cana listings, which justifies using regression to explain that variation.

Price vs. Bedrooms

```
ggplot(airbnb, aes(bedrooms, price)) +  
  geom_point(alpha = 0.6, color = "darkred") +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Nightly Rate vs. Bedrooms",  
       x = "Bedrooms", y = "Nightly Rate")
```

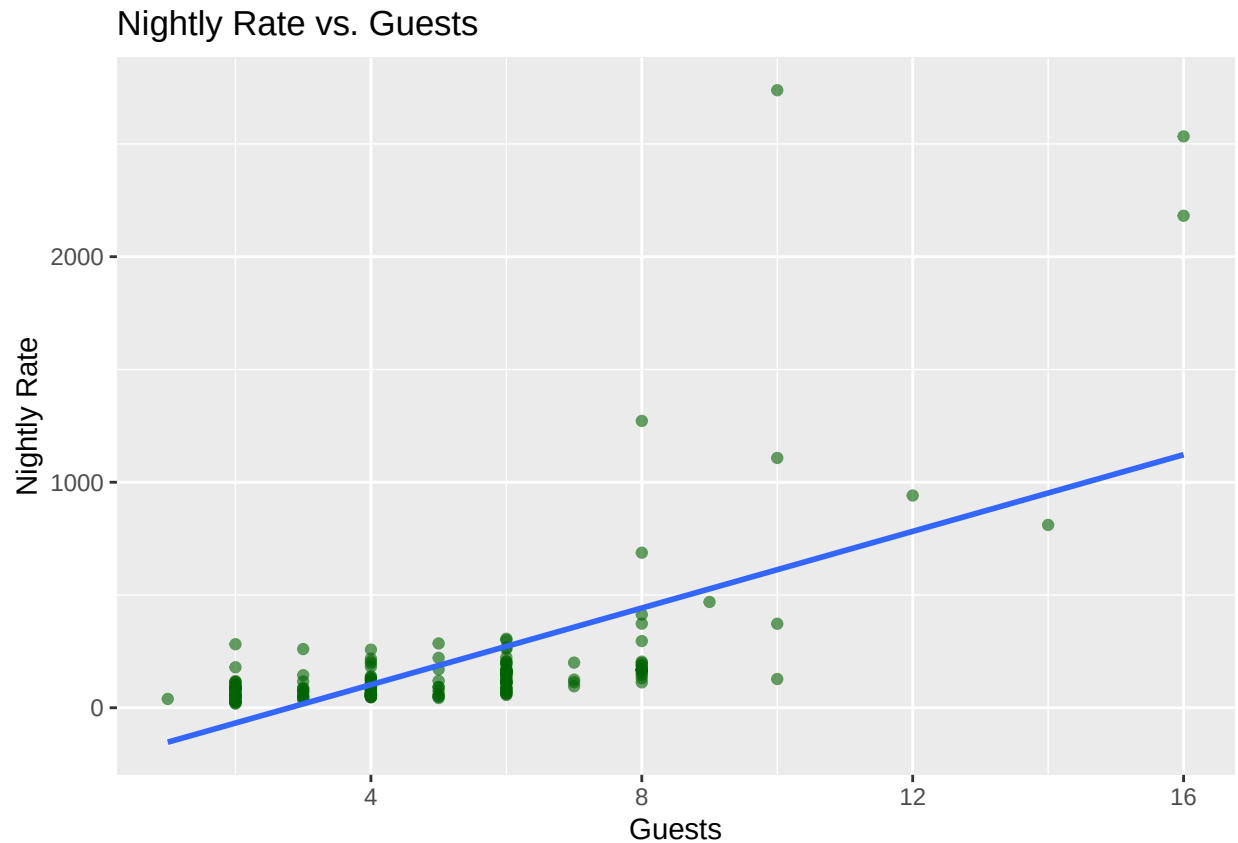


Commentary:

This scatterplot shows a positive relationship between the number of bedrooms and nightly price. Listings with more bedrooms tend to charge higher rates. The fitted regression line reinforces this upward trend, indicating that property size is an important driver of price.

Price vs. Guests

```
ggplot(airbnb, aes(guests, price)) +  
  geom_point(alpha = 0.6, color = "darkgreen") +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Nightly Rate vs. Guests",  
       x = "Guests", y = "Nightly Rate")
```

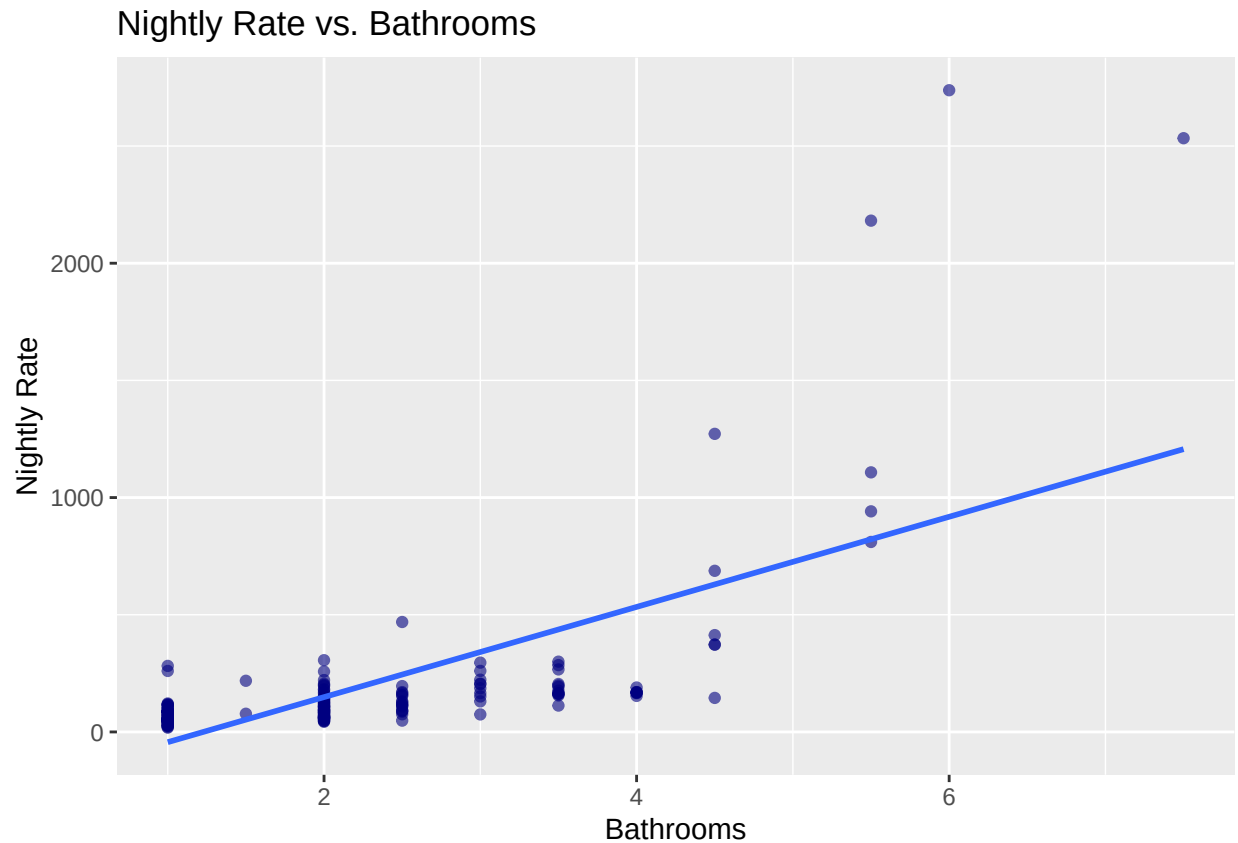


Commentary:

This scatterplot shows that listings accommodating more guests generally have higher nightly prices. This makes sense because higher guest capacity usually corresponds to larger or more desirable properties. The fitted line shows a clear upward trend.

Price vs. Bathrooms

```
ggplot(airbnb, aes(baths, price)) +  
  geom_point(alpha = 0.6, color = "navy") +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Nightly Rate vs. Bathrooms",  
       x = "Bathrooms", y = "Nightly Rate")
```



Step 5: Correlation Analysis and Variable Selection

```
airbnb_num <- airbnb |>
  select(price, guests, bedrooms, baths)

cor(airbnb_num)
```

```
##           price    guests bedrooms    baths
## price      1.000000 0.6388722 0.6385465 0.6769204
## guests     0.6388722 1.0000000 0.9276069 0.8768011
## bedrooms   0.6385465 0.9276069 1.0000000 0.9334578
## baths      0.6769204 0.8768011 0.9334578 1.0000000
```

Commentary:

- All three predictors show positive correlations with price.
- These variables were selected because they are numeric, interpretable, and logically related to property size.

Step 6: Multiple Regression Model (3 Predictors)

```
model1 <- lm(price ~ guests + bedrooms + baths, data = airbnb)
summary(model1)

##
## Call:
## lm(formula = price ~ guests + bedrooms + baths, data = airbnb)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -449.00 -112.76  -39.04  100.17 1867.08
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -255.52      39.37  -6.490 7.56e-10 ***
## guests         38.09      19.07   1.997  0.0473 *
## bedrooms     -56.93      57.27  -0.994  0.3215
## baths        171.72      42.44   4.046 7.62e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 245.3 on 186 degrees of freedom
## Multiple R-squared:  0.4699, Adjusted R-squared:  0.4614
## F-statistic: 54.97 on 3 and 186 DF,  p-value: < 2.2e-16
```

How much of price does the model explain?

R-squared = 0.4699

This means the model explains **about 47% of the variation** in nightly price.
For a simple model using only three numeric predictors, this is a strong result.

Step 7: Regression Equation and Interpretation

```
coef(model1)
```

```
## (Intercept)      guests      bedrooms      baths
## -255.52127    38.08731   -56.93020    171.71981
```

$$\text{price} = \beta_0 + \beta_1(\text{guests}) + \beta_2(\text{bedrooms}) + \beta_3(\text{baths})$$

Interpretation (Addressing Overlapping Predictors)

- Guests, bedrooms, and baths all show **positive** relationships with price in the EDA.
- However, in the regression model, **bedrooms becomes negative**.
- This does **not** mean that adding a bedroom lowers the price.

Why did bedrooms turn negative?

Guests and bedrooms both measure **property size**, and larger properties tend to increase all three variables together.

Because these predictors overlap, the model struggles to separate their individual effects. This overlap is called **multicollinearity**.

Note on Googling VIF

I looked up why coefficients flip signs and learned that the recommended diagnostic is a **VIF test**, but VIF is **outside the scope of this 300-level course**, so I did not run it. Instead, I relied on the tools covered in class:

- scatterplots
- correlation matrix
- regression output

Final interpretation

The negative bedrooms coefficient is a **statistical artifact** of overlapping predictors, not a real-world effect.

The overall conclusion remains clear: **larger, higher-capacity properties tend to have higher nightly rates**.

Step 8: Model Diagnostics (Removed)

R automatically produces diagnostic plots for linear regression models.

These diagnostics are typically covered in more advanced courses, so they were **not included or interpreted** for this project.

Key Findings

1. Listings with more guests, bedrooms, and bathrooms tend to have higher nightly rates.
2. A simple three-predictor model explains a meaningful portion of price variation ($R^2 = 47\%$).
3. The negative bedrooms coefficient is due to overlapping predictors, not a real effect.

Limitations

- Observational data cannot establish causality.
- Important factors such as amenities, location, and host ratings are not included.
- Results apply only to Punta Cana.

Conclusion

This project applied introductory regression techniques to real Airbnb data. Following a clear step-by-step process, I:

1. Loaded and cleaned the dataset
2. Identified numeric variables
3. Performed EDA
4. Selected guests, bedrooms, and baths as predictors
5. Fit a multiple regression model
6. Interpreted the model, including overlapping predictors
7. Summarized findings and limitations

The results show that property size and capacity are key drivers of nightly rates in Punta Cana.